PRISMA 2020 Eligibility Stage: Quality Criteria Scoring — Section 4.3 (Drift Benchmarking and Controlled Drift Injection)

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1. Scope of This Scoring Pass

This report extends the Step 4 eligibility scoring to the full candidate set retrieved for Section 4.3 (Drift Benchmarking and Controlled Drift Injection) of the review, comprising all 9 full-text articles retained for this subsection following Step 3 screening. Each article was scored honestly against its full text rather than against title and abstract alone.

QC1 (problem statement and scope clarity) was scored against fit to Section 4.3's specific review question, namely whether an article's core contribution is a benchmarking-methodology critique, an open-source synthetic drift-injection framework or generator, or a documented real-world drift-dataset repository, rather than against the clarity of the problem statement in the article's own, potentially different scope. An article can present an impeccably clear problem statement for a different question, such as proposing a new drift-detection algorithm, or benchmarking detectors on a non-IoT application domain, and still score low on QC1 here, because that problem statement is not the one this subsection screens for.

QC4 (empirical extraction detail) was scored as 0 wherever an article's full text contained no drift-type, distributional-component, or adaptation-mechanism data extractable into this review's own extraction fields, regardless of how much extraction detail the article offered for its own, different research question. This reflects a consistent, review-specific application of the criterion rather than a per-article adjustment.

QC2 (methodological transparency) and QC3 (venue indexing and citability) were scored on their ordinary, article-intrinsic basis, since these criteria measure the rigor and citability of the work itself rather than its fit to this particular review.

2. Quality Assessment Criteria Applied

Table 1 reports the four quality criteria (QC1 to QC4) applied in this eligibility round, together with the scoring scheme used for each. A quality score is computed as the sum of applicable criterion scores divided by the number of applicable criteria, scaled to a 0 to 4 range.

Table 1. Quality assessment criteria applied to Section 4.3 candidate articles

Code	Criterion	Scoring
QC1	Problem statement and scope clarity	Yes (1) / Partial (0.5) / No (0)

Code	Criterion	Scoring
QC2	Methodological transparency	Yes (1) / Partial (0.5) / No (0)
QC3	Venue indexing and citability	Yes (1) / Partial (0.5) / No (0)
QC4	Empirical extraction detail (where applicable)	Yes (1) / Partial (0.5) / No (0) / N/A

Articles published in 2020 or later were retained where their total quality score met or exceeded 2.5 out of 4. Articles published before 2020 were retained where their total quality score met or exceeded 1.5 out of 4.

3. Results

Of the 9 full-text articles assessed for Section 4.3 against the quality criteria in Section 2, 5 articles met or exceeded their applicable threshold and were retained, while 4 articles fell below their applicable threshold and were excluded. Table 2 lists each assessed article together with its per-criterion and total quality scores, applicable threshold, and eligibility decision, sorted with included articles listed first, followed by excluded articles.

Table 2. Quality criteria scoring results for Section 4.3 candidate articles, sorted by decision

No.	Citation	Year	QC1	QC2	QC3	QC4	Total	Threshold	Decision
1	MOA: Massive Online Analysis (2010)	2010	1.0	1.0	1.0	1.0	**4.0**	≥1.5	**Include**
2	Stream-learn - open-source Python library for difficult data stream batch analysis (2022)	2022	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
3	Challenges in benchmarking stream learning algorithms with real-world data (2020)	2020	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
4	Synthetic non-stationary data streams for recognition of the unknown (2026)	2026	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
5	Concept Drift Detection in Data Stream Mining: A literature review (2022)	2022	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
6	DRIFTNET-EnVACK: Adaptive Drift Detection in Cloud Data Streams With Ensemble Variational Auto-Encoder Featuring Contextual Network (2024)	2024	0.0	1.0	1.0	0.0	**2.0**	≥2.5	**Exclude**
7	Concept drift detection methods based on different weighting strategies (2024)	2024	0.0	1.0	1.0	0.0	**2.0**	≥2.5	**Exclude**
8	Concept anchor drift detection: a framework for proactive detection of abrupt drifts in real-world data streams (2026)	2026	0.0	1.0	1.0	0.0	**2.0**	≥2.5	**Exclude**

No.	Citation	Year	QC1	QC2	QC3	QC4	Total	Threshold	Decision
9	Concept drift detection in district heating: A characterization framework and detector-centric benchmark (2026)	2026	0.5	0.5	1.0	0.0	**2.0**	≥2.5	**Exclude**

This retained set of 5 articles constitutes the eligible pool for Section 4.3 (Drift Benchmarking and Controlled Drift Injection) and forms part of the input to the synthesis and thematic analysis stage.

4. Data Integrity Notes

Each article was scored against the four quality criteria on the basis of its full text, applying the review-specific interpretation of QC1 and QC4 set out in Section 1 above. The retain or exclude decision was applied strictly according to the threshold in effect for that article's publication year, namely 2.5 out of 4 for articles published 2020 or later, or 1.5 out of 4 for articles published before 2020. No article was retained or excluded on any basis other than its quality score relative to its applicable threshold.

5. Document Preparation

This report has been prepared in accordance with the quality assurance procedure prior to proceeding to the synthesis stage.

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